

Magneto: Enabling Multimodal Haptic Feedback on Paper through Magnetic Fields

Jeongju Park
Human-Centered Intelligent Systems
Lab
Gwangju Institute of Science and
Technology
Gwangju, Republic of Korea
jeongjupark@gm.gist.ac.kr

Semoo Shin
Human-Centered Intelligent Systems
Lab
Gwangju Institute of Science and
Technology
Gwangju, Republic of Korea
sms081700@gm.gist.ac.kr

Seongjun Kang
Human-Centered Intelligent Systems
Lab
Gwangju Institute of Science and
Technology
Gwangju, Republic of Korea
ksjryan0728@gm.gist.ac.kr

Gwangbin Kim
Human-Centered Intelligent Systems
Lab
Gwangju Institute of Science and
Technology
Gwangju, Republic of Korea
gwangbin@gm.gist.ac.kr

SeungJun Kim*
Human-Centered Intelligent Systems
Lab
Gwangju Institute of Science and
Technology
Gwangju, Republic of Korea
seungjun@gist.ac.kr

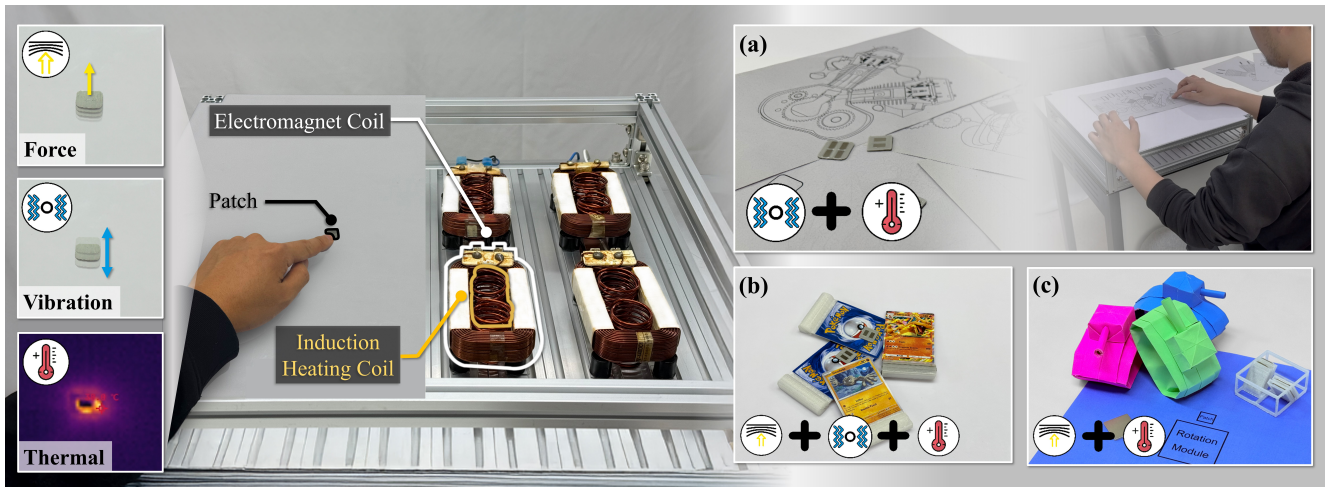


Figure 1: Overview of the Magneto system showing actuation modules with electromagnetic and induction heating coils enabling force, vibration, and thermal feedback, (a) integration in technical drawings with vibration and thermal feedback, (b) card game with force, vibration, and thermal actuation, and (c) origami prototype involving force and thermal feedback capabilities.

Abstract

Paper, with its inherent versatility and adaptability, has long served as an accessible medium for interaction across different use cases. However, its analog nature constrains it to relatively fixed, static

*Corresponding author.

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uses once created and limits its ability to provide richer sensory feedback or respond to changing contexts in the way digital interfaces can. To address this limitation, we introduce Magneto, a multimodal haptic system that augments paper interfaces with force, vibration, and thermal feedback through magnetic fields. The system combines neodymium magnets and steel plates in patch modules, using electromagnets for force and vibration and Zero Voltage Switching (ZVS) induction modules for thermal feedback. The technical evaluation confirms Magneto's operation within human perceptual thresholds across the suggested feedback modalities. Applications of Magneto are explored in educational, gaming, and prototyping contexts, illustrating its potential to transform static paper interfaces into dynamic, interactive systems.

CCS Concepts

• Human-centered computing → Haptic devices.

Keywords

Paper Interface, Haptic, Magnetic-based, Multimodal, Multisensory

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1 Introduction

Paper has long served as a versatile medium for transmitting information through text and images, constructing three-dimensional structures through folding, and supporting educational tools such as flashcards [32] and entertainment applications including game cards [12, 32]. Its simplicity and adaptability allow for various customizations and fabrications, including cropping, printing, folding, and embossing, making it a widely used material for human expression and interaction [4, 15]. However, traditional paper remains analog, its interactions are confined to a narrow range of sensory experiences, centered solely on visual perception [9].

Meanwhile, as mobile and interactive technologies become increasingly prevalent and embedded, they incorporate sensory feedback mechanisms like force feedback, vibration, and thermal variations [10, 11, 13, 14]. These features foster user immersion and engagement by enriching sensory interaction with information beyond static presentation [25]. Despite paper's widespread utility, it lacks the adaptability to provide dynamic feedback beyond its physical properties [22].

In response to this limitation, methods for augmenting the paper's interactive potential have emerged. Shape memory alloys (SMAs) [16, 28, 30, 35, 47, 48] and conductive paint [19, 37, 38] enable shape-changing and thermal feedback, enhancing engagement in educational activities [30, 31] and creative tasks such as origami [22]. While these approaches demonstrate potential for multimodal paper interfaces, their reliance on complex wiring, power supplies, and actuation systems compromises paper's inherent simplicity and multifunctionality [24].

To address these challenges, this study introduces Magneto, a multimodal haptic system that enhances paper interfaces through patch-shaped ferromagnetic materials (Fig. 1). Using external magnetic fields, Magneto delivers force, vibration, and thermal feedback via a single, hardware-efficient mechanism. This approach preserves paper's lightweight nature while enabling interactive capabilities by attaching modular patches without additional modifications. Through case studies in education and entertainment, we demonstrate how Magneto expands the possibilities for paper interfaces while maintaining their advantages, such as versatility and sensory richness [2].

2 Related Work

2.1 Paper Interface

Despite the pervasiveness of digital devices such as computers and tablets in our modern lives, paper remains a widely used medium for conveying information and enabling creative expression, as evidenced by its use in pop-up books [29], origami [30], and handmade crafts [35]. Its low cost, immediate availability, and tangible nature provide unique advantages that digital alternatives have not fully replaced, particularly in educational settings, creative processes, and professional environments where quick notation, sketching, and crafting are valued. Its versatility allows for a wide range of modifications, including cropping, printing, folding, and embossing, making it a flexible platform for diverse applications. To enhance paper's interactivity, researchers have explored methods to augment it with haptic feedback, spanning applications in games [12, 32], education [30, 31] art [5, 29] and prototyping [25, 35, 39].

Early approaches demonstrated ways to introduce dynamic physical changes to paper interfaces. Sticky Actuator [22] implemented a tape-based pneumatic system that enabled shape-changing interfaces through air pressure modulation during origami activities. Other studies explored force feedback through shape memory alloys (SMAs) [16, 28, 30, 35, 47, 48] and shape memory polymers (SMPs) [25, 39], while Move-it Sticky Notes [28] combined SMA clips with Post-it notes to create tangible notification devices. While these interfaces provide dynamic, shape-changing properties to paper, they sometimes had to compromise the size and complexity of paper interfaces through integrated circuits [28, 30, 35], wiring [16, 25, 39], or pneumatic tubes [22].

Research has also explored tactile feedback through vibration and thermal sensations. CARDinality [32] integrated vibration actuators and communication modules into paper cards, while Painting Inferno [37] utilized carbon-based heating paint for thermal feedback. While these approaches achieved tactile feedback through compact designs, they relied on specific circuit configurations [32] and material properties [37], which sometimes constrained applications to fixed components and required specific preconditions. In this work, we explore modular patches that can augment plain paper with force, vibration, and thermal feedback capabilities without permanent modifications, aiming to maintain paper's inherent flexibility while enabling haptic interactions.

2.2 Magnetic-Based Interface

Magnetic field-based methods offer a way to provide force feedback through external magnetic field control without internal power sources [40, 41]. This approach reduces reliance on wiring and complex circuitry through non-contact actuation [43, 45]. These methods have been applied to paper interfaces [6, 24, 42, 46] as well as everyday objects, including walls [20], cups [27], and blocks [23], demonstrating broad applicability across different materials and contexts.

Research has explored various implementations of magnetic actuation. FluxPaper [24] enabled paper movement using thin neodymium patches, demonstrating applications in Post-it notes and origami. MagPixel [8] utilized magnetically soft materials to create interactive shape-changing displays, with applications including DIY tangible clocks and reminders. MagnetIO [20] explored

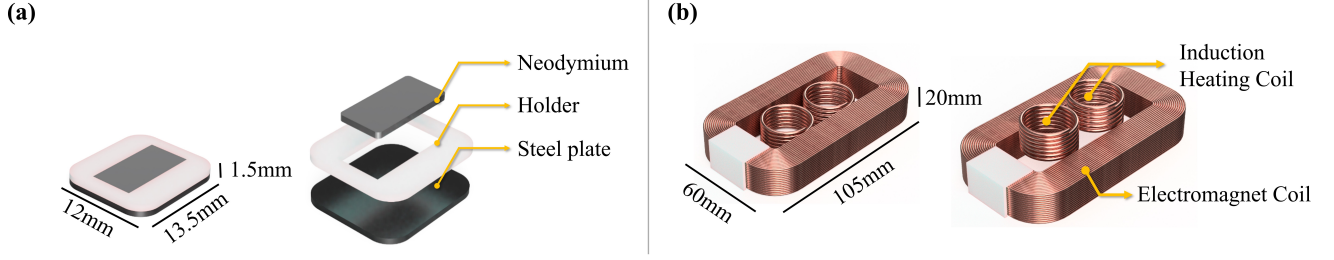


Figure 2: Hardware components showing (a) a bottom-up view of the structural breakdown of a Magneto patch composed of steel plate, neodymium magnet, and holder, and (b) electromagnetic and induction coil modules.

wearable, finger-sized coils that interact with soft magnetic patches attached to everyday objects, providing vibration feedback for enhanced interactivity. Similarly, Magnetact [44] combined patterned magnetic sheets with conductive materials for force and vibration feedback, while Magneto-Haptics [23] used angled permanent magnets within blocks to generate a set of haptic patterns.

While these approaches effectively deliver force and vibration feedback without mechanical actuators, the low electrical conductivity of neodymium magnets limits thermal actuation capabilities [17, 34]. Current systems require additional components such as Peltier modules for thermal feedback [1], which increases system complexity and compromises the simplicity of magnetic-based mechanisms. In this work, we explore the combination of magnetic field-based approaches to enable force, vibration, and thermal feedback through a unified mechanism, aiming to expand the range of haptic interactions while maintaining system simplicity.

3 Implementation

Magnetio consists of an actuating module, which includes an electromagnet and a Zero Voltage Switching (ZVS) induction heating component, and a patch composed of neodymium magnets and steel plates. This system delivers multimodal haptic feedback through magnetic field variations, providing force, vibration, and thermal sensations via dedicated patches.

3.1 Patch Design

As shown in Fig. 2 (a), a single Magneto patch consists of a permanent magnet layer made from a $12\text{ mm} \times 13.5\text{ mm} \times 0.7\text{ mm}$ steel plate and a $5\text{ mm} \times 10\text{ mm} \times 0.8\text{ mm}$ neodymium (Nd-Fe-B) magnet, which is held securely in place by a 3D-printed holder. The entire patch weighs approximately 1 g, ensuring minimal impact on the paper interface. The neodymium magnet generates attractive and repulsive forces through an external magnetic field, enabling force and vibration feedback. While neodymium provides strong magnetic coupling for mechanical feedback, its low electrical conductivity makes it less suitable for induction heating. To address this limitation, the steel plate's high electrical conductivity enables efficient thermal feedback through eddy current and hysteresis losses during induction heating. The steel plate is attached directly to the paper surface in contact with the user, minimizing heat loss during thermal feedback. This complementary combination of materials

in the patch design enables Magnetio to deliver force, vibration, and thermal feedback through a single unified structure.

3.2 Actuating Module

The actuation module comprises an electromagnet module for force and vibration feedback and a ZVS induction heating module for thermal feedback (Fig. 2 (b)). A Teensy 4.1 microprocessor independently controls each module, enabling simultaneous multimodal feedback across multiple patches.

The control system generates PWM signals for the electromagnetic module and binary signals for the ZVS induction module. Amplified by a TLP250 gate driver, these signals control an IRG4PC50WPBF IGBT, switching high-voltage and high-current loads to create the required magnetic fields. This configuration allows control over the magnetic field characteristics for each feedback type. The electromagnetic module ($60\text{ mm} \times 105\text{ mm} \times 20\text{ mm}$) operates at a supply voltage of 24V and consumes up to 16A, reaching a maximum power consumption of 390W, providing sufficient force and vibration range, while the ZVS induction module's compact coil design ($25\text{ mm} \times 25\text{ mm} \times 20\text{ mm}$) begins resonant operation when drawing 12V at 10A, consuming 120W (lower than the 1,200–2,500W typical of household induction cooktops), which enables the generation of localized thermal feedback. The ZVS induction module generates a high-frequency magnetic field of approximately 30 kHz, with the exact frequency determined by the capacitance and inductance of its resonant circuit. The modular design enables independent adjustment of each feedback parameter while maintaining synchronized operation when multiple feedback types are required.

3.3 Feedback Mechanisms

Force Feedback via Duty Cycle Modulation Force feedback is achieved by modulating the PWM signal's duty cycle. This modulation controls the average current to the electromagnet, varying the magnetic field strength [26]. This allows Magnetio to regulate the repulsive force applied to the patch, providing force feedback.

Vibration Feedback via Frequency Modulation Vibration feedback is implemented by modulating the frequency of the PWM signal. By altering the switching frequency of the electromagnet, the system creates periodic magnetic fields that apply repetitive repulsive forces to the patch, generating controlled vibrations.

Thermal Feedback via High-Frequency Bipolar Signal Modulation Thermal feedback utilizes a ZVS induction heating module

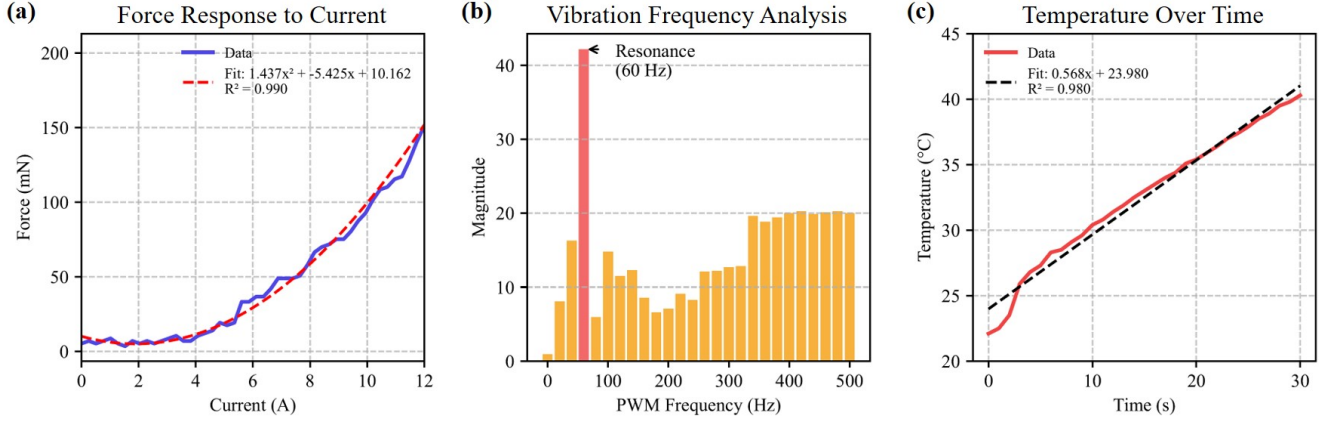


Figure 3: Technical evaluation results showing (a) force output relative to current input, (b) magnitude values obtained from the piezoelectric sensor as a function of PWM frequency, with peak at 60 Hz, and (c) thermal response demonstrating linear temperature increase over time.

comprising an LC resonant circuit and MOSFETs. The module converts DC power into high-frequency switching signals through the LC resonant circuit, which controls the MOSFET gates. These signals are amplified by the LC resonant circuit to produce high-frequency bipolar signals. The resulting magnetic field induces eddy current and hysteresis losses within the patch material, generating controlled heat for thermal feedback. Following the heating phase, the patch undergoes passive cooling primarily through natural convection, where heat is transferred to the surrounding environment until the patch temperature stabilizes at ambient conditions.

4 Evaluation

To quantitatively evaluate the feedback performance of Magneto, we conducted experimental tests using a 12 mm × 13.5 mm one-cell patch. This evaluation includes measurements of (1) force feedback using a load cell, (2) vibration feedback using a piezoelectric vibration sensor, and (3) thermal feedback using an infrared thermal camera.

Force Feedback To measure the force output of the Magneto patch, we used a CLMN-5L load cell positioned 3mm above the patch in a 3D-printed holder. The aluminum construction minimized electromagnetic interference, and the holder stabilized the patch to prevent X-Y movement during measurements. The patch was centrally aligned with the electromagnetic module, maintaining a vertical gap of 3mm. The current supplied to the electromagnet was incrementally increased from 0A to 12A by modulating the duty cycle of the PWM signal, with data recorded at 10Hz.

Fig. 3 (a) shows the repulsive force applied to the patch as a function of current. The force increased along a quadratic curve, reaching a maximum of 147mN. Meanwhile, the detection threshold for force perception through the fingertips' glabrous skin is 50 mN [7], indicating that Magneto can provide users with perceivable force feedback. This capability can be further enhanced by combining multiple one-cell patches, enabling perceptible haptic feedback across interaction scenarios.

Vibration Feedback We measured vibration feedback using a piezoelectric vibration sensor (KY0138). The patch was positioned at the center of the electromagnetic module with a 3 mm gap, and vibration feedback was measured while varying the PWM signal frequency from 0 Hz to 500 Hz in 10 Hz steps. Fig. 3 (b) shows the patch's magnitude response, which represents the raw analog signal directly acquired from the piezoelectric sensor to the electromagnetic module's magnetic field frequency. The system exhibited resonance at approximately 60 Hz. However, this resonance frequency is not fixed and may vary based on the weight of attached components. As weight increases, the resonance frequency decreases, while lighter attachments result in a higher resonance frequency [33].

Vibration feedback was detected across the entire 0–500 Hz range, covering the human vibrotactile perception range of 40–500 Hz [21]. Specifically, human tactile perception in the glabrous skin involves four mechanoreceptive channels that collectively process signals within 0.4–500 Hz [3]. Merkel's disks process 0.4–2.0 Hz signals as "pressure", Meissner corpuscles detect 2–40 Hz as "flutter", Pacinian corpuscles respond to 40–500 Hz vibrations, and Ruffini end organs interpret 100–500 Hz as "buzzing" [21]. The frequency band of Magneto aligns with these ranges, enabling feedback across mechanoreceptor sensitivities to provide pressure, flutter, and buzzing sensations through patch vibration.

Thermal Feedback We measured thermal feedback using a thermal camera (Lepton 3.5). The patch was positioned 3 mm above the ZVS induction heating coil, with activation set to heat the patch to 40°C, a temperature threshold chosen to ensure safe yet perceivable thermal feedback. Temperature measurements were recorded at a sampling rate of 1 Hz.

Fig. 3 (c) illustrates the patch's temperature response to the induction module's high-frequency magnetic field. The temperature rose linearly from 22°C to 40°C over 30 seconds (0.6°C/s), exceeding the human thermal response threshold of 0.1°C/s [1]. This ensures perceptible temperature changes within human thermal perception ranges. Additionally, human skin perceives mild to warm sensations

through TRPV4 (27–34°C) and TRPV3 (32–39°C) sensory receptors [18]. Magneto’s heating capability spans these physiological ranges, enabling activation of both receptor types through controlled, consistent thermal feedback.

5 Applications

Based on the evaluation results demonstrating alignment with human perceptual thresholds, we explore Magneto’s applications across different scenarios. We explore how multimodal haptic feedback can be incorporated into paper-based practices. By combining patches with various geometric configurations, the system extends force feedback into yaw and pitch rotations beyond simple paper interfaces. We examine implementations in productivity, gaming, and prototyping applications.

5.1 Augmented Drawing with Haptic Feedback

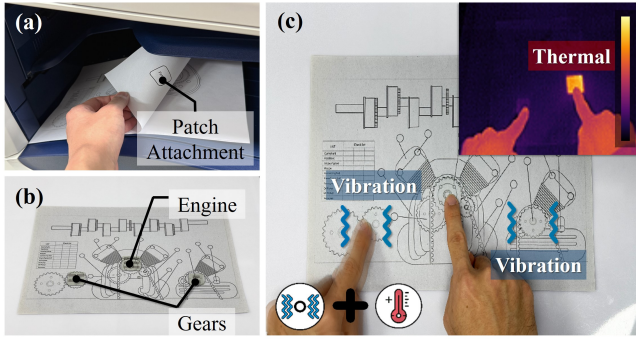


Figure 4: Technical drawing application with (a) double-sided printing of patch locations, (b) diagram with marked interaction points, and (c) multimodal feedback simulating component behaviors with vibration and thermal sensations.

Magneto augments traditional paper interfaces in office scenarios through multimodal haptic feedback. Fig. 4 demonstrates this capability in technical drawings and blueprints, where embedded patches provide contextual feedback without requiring additional wiring or circuitry. The patch locations are printed on the back of the paper using double-sided printing (Fig. 4 (a)), and the patches are attached to the corresponding positions on the back of the paper (Fig. 4 (b)). Through this setup, the system provides haptic sensations such as vibration or thermal feedback, simulating functional characteristics of illustrated components, such as engine vibrations or thermal variations from power systems (Fig. 4 (c)). This approach enables an intuitive understanding of the depicted system’s operation.

5.2 Interactive Card Game

Magneto introduces interactive elements to card games, such as monster battle card games like Pokémon or Yu-Gi-Oh!, by enabling haptic feedback during gameplay. By attaching Magneto patches to the back of cards and incorporating 3D-printed rotational modules, cards can deliver vibration, thermal, and rotational feedback during play.

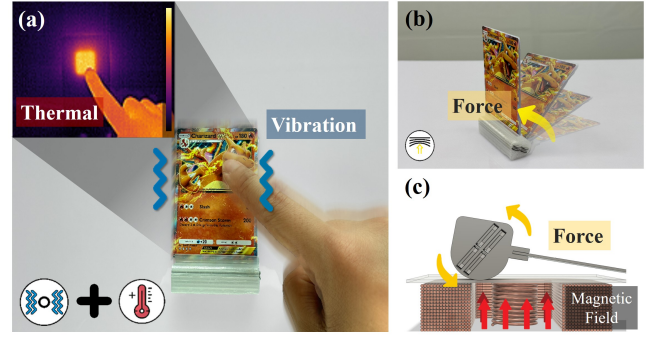


Figure 5: Interactive card game demonstrating (a) thermal and vibration feedback for a fire-type card, (b) force feedback for card movement, and (c) mechanism diagram showing magnetic field interactions for card actuation.

Fig. 5 illustrates the use of Magneto in a Pokémon card game. Players attach a Magneto patch to the back of their cards and place the cards into a 3D-printed roll rotation module. For instance, when playing a fire-type card, the Magneto patch provides thermal and vibration feedback to simulate the heat and intensity of a fire attack (Fig. 5 (a)). When a player declares an attack, the card pops up visually using the pitch rotation module to indicate the attack action (Fig. 5 (b)). The mechanism behind this action is visualized in Fig. 5 (c), showing how the pitch rotation module activates and then deactivates the magnetic field to allow the card to return to its original position.

The pitch rotation module was designed based on the experimental data in the Evaluation section. Even when the distance between the electromagnet and the patch was 10mm and the effective magnetic field strength was reduced to approximately 9% due to a 60° tilt, the module actuated a 7.8g card (including the Magneto patch for vibration and thermal feedback). This was achieved by distributing 16 one-cell patches over a 17mm × 65mm area.

5.3 Prototyping

Magneto can be used in prototyping processes involving origami by adding haptic feedback capabilities. While traditional approaches to origami-based prototyping either had limited haptic feedback to showcase functions or require complex wiring and circuitry for deformation, vibration, or thermal sensations that limited flexibility, Magneto’s embedded patches can deliver multimodal feedback without additional wiring.

Fig. 6 demonstrates this capability through an interactive tank origami prototype using Magneto. Fig. 6 (a) shows a printed paper template that specifies predefined locations for attaching Magneto patches and modules. The components are attached to the designated positions inside the origami structure during the folding process. The rear patch of the tank is designed to simulate heat generation from the engine, adding a layer of realism to the prototype. The completed tank features a yaw rotation module integrated into its structure, enabling rotational motion. Fig. 6 (b) illustrates the tank equipped with embedded modules rotating as controlled

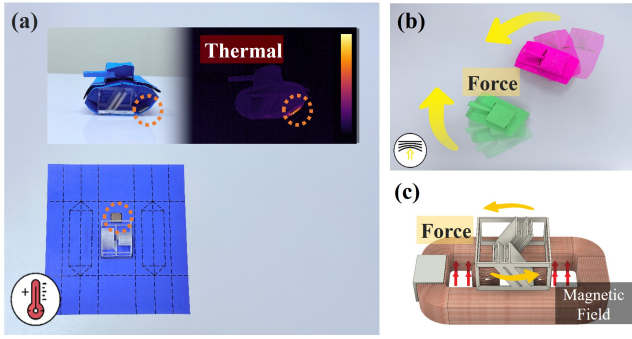


Figure 6: Interactive origami prototype showing (a) thermal imaging and patch placement guide, (b) force-based actuation of folded components, and (c) diagram illustrating interaction between magnetic fields and patches to achieve yaw rotation.

by the yaw rotation module. Fig. 6 (c) visualizes the underlying mechanism of this rotation module.

Due to the increased distance of 8mm between the electromagnet and the patch, which is approximately 2.67 times the 3mm distance used in the Evaluation section, the effective force was reduced to approximately 14.1%. To compensate, two additional one-cell patches were placed on either side of the rotation axis, totaling four patches. The patches were positioned at 60° angles relative to the rotation axis, ensuring a perpendicular magnetic field interaction with their side surfaces. This configuration enables the magnetic force to induce rotational movement, creating a functional yaw rotation mechanism. As the patches rotate, the magnetic flux acts more directly on their surfaces, enabling consistent rotational force and enhancing the overall interactivity of the origami design.

6 Discussion

Performance Variations Based on Patch Position Our performance evaluations were conducted with the patch positioned at the center of the electromagnet. In real applications, the magnetic flux density and field direction may vary depending on the patch’s position relative to the electromagnet, potentially affecting system performance. Future work should systematically investigate how these positional variations influence the patch’s force, vibration, and thermal feedback performance. Additionally, exploring different neodymium magnet configurations within patches, including their position and orientation, could expand the potential design space and enhance the system’s versatility.

Development of Flexible Patches Using Powder Mixtures The current Magneto patch uses rigid steel plates and neodymium magnets, limiting flexibility. Previous studies on magnetic powder mixtures [8, 20, 24, 36] suggest the potential of creating flexible patches suitable for both paper and curved surfaces. Incorporating neodymium (Nd-Fe-B) powder mixed with iron powder could enable flexible patches with optimized properties. While adding iron powder might reduce magnetic strength, the resulting changes in electrical conductivity could influence the force, vibration, and

thermal feedback characteristics. Exploring the optimal composition of neodymium and iron. Future research could explore optimal powder mixture ratios to develop flexible patches that perform effectively across their use cases.

7 Limitation and Future work

User Study Our work focused on the technical foundations of Magneto through characterization of its hardware specifications and demonstrated application scenarios that leverage the system’s force, vibration, and thermal haptic capabilities. While the technical evaluation helped understand the system’s characteristics and the demonstrations illustrated possible integration pathways, future work could enrich these findings through structured user studies.

Power Consumption Our work enhances haptic feedback in paper interfaces through electromagnetic actuation without requiring complex design modifications, aligning with paper’s inherent adaptability for customization. However, a limitation of our approach is the high power consumption required by the system, consuming 390W for force and vibration feedback and 120W for induction heating. While this design was intended to generate a strong magnetic field to produce high forces through patches, its energy requirement poses challenges for energy efficiency, portability, and extended use scenarios where power use would be limited.

Future research could explore optimizing power efficiency by increasing the number of coil windings while reducing the thickness of the wire used in the electromagnet, thereby lowering current draw without compromising actuation performance. This approach is expected to maintain multimodal haptic feedback while achieving a more energy-efficient design.

Workspace Constraints Magneto employs magnetic field-based actuation, which imposes constraints on its physical workspace. The magnetic field-generating coils are fixed within a non-portable frame structure, restricting the system’s applicability to tabletop environments. This fixed nature limits deployment flexibility and prevents usage in mobile or space-constrained contexts.

To address these spatial limitations, future research should explore reducing the size of the actuating modules. By designing actuating modules to be more compact and lightweight through miniaturized electromagnetic components and streamlined ZVS induction heating systems, individual modules could be made portable, allowing them to be independently repositioned and deployed in different configurations according to application needs.

Resolution Within the magnetic field range of a single actuating module, patches are difficult to control independently. This limitation restricts the ability to provide differentiated haptic feedback to multiple paper elements located within the same module’s magnetic field, constraining the granularity of interactions possible within a small area. Miniaturizing the actuation modules could also enhance the resolution of haptic feedback. Smaller modules would create more localized magnetic fields, enabling finer control over individual patches and more precise tactile experiences.

8 Conclusion

This paper introduced Magneto, a device that enhances paper interfaces with multimodal haptic feedback through a unified magnetic field-based mechanism. The system delivers force, vibration, and

thermal sensations using patches composed of steel plates and neodymium magnets in conjunction with electromagnetic and induction modules. Force and vibration feedback are achieved through interactions between the patches' neodymium magnets and the electromagnetic module, while thermal feedback is generated through the induction module's interaction with the steel plates. As one of the first explorations of magnetic field-based thermal feedback for interaction, this approach demonstrates new possibilities for augmenting paper interfaces with rich haptic capabilities.

Through technical evaluation and demonstrations across productivity, gaming, and prototyping contexts, we showed how MagnetO's unified approach to multimodal haptic feedback enables customizable tactile experiences. The system's ability to transform everyday materials into interactive interfaces suggests potential applications in paper-based interface design. Future research could explore flexible patch materials and systematic investigation of positional effects to expand the system's application domains.

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